Campus Information Retrieval Assistant Using Small Language Model

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INTRODUCTION

- Universities generate vast amounts of information across multiple sources such as websites, PDFs, and notices
- Students and stakeholders often struggle to access relevant information quickly
- Traditional search methods are inefficient and time-consuming
- AI-based assistants can significantly improve information accessibility
- This project introduces a campus-specific intelligent assistant

PROBLEM STATEMENT

- Campus information is highly fragmented across different platforms
- Users spend significant time searching for simple queries
- Existing chatbot systems rely on cloud-based LLMs which cause:
  - High operational costs
  - Privacy and data security risks
  - Hallucinated or inaccurate responses
- Lack of systems prioritizing institutional data sources

Therefore, a reliable, privacy-focused, campus-specific solution is required

OBJECTIVES

- Design a Small Language Model (SLM)-based assistant
- Provide accurate and fast responses
- Minimise hallucinations using RAG
- Enable on-premise deployment
- Prioritise institutional data sources
- Support natural language queries

LITERATURE REVIEW

Title	Author(s)	Year	Published Area / Site	What They Did	How It Is Different From Our Work	Link to Paper
An Empirical Study of Multi-Agent RAG for Real-World University Admissions Counseling	Anh Nguyen-Duc	2025	arXiv (cs.IR, cs.SE)	Built MARAUS, a multi-agent RAG system using LLMs for university admissions counselling; evaluated on 6,000+ real queries with strong accuracy and low hallucination rates.	Uses cloud LLMs and multi-agent orchestration focused only on admissions; our system uses a single SLM, broader campus-wide information, strict source prioritisation, and on-prem/offline deployment.	https://arxiv.org/abs/2507.11272
Design of a RAG-based Q&A System for University Academic Administration Assistants	Jae-Kwang Oh, Soo-Kyun Kim	2025	Journal of the Korea Society of Computer and Information	Designed a RAG-based Q&A assistant to support academic administrative queries within universities.	Focuses mainly on administrative workflows and traditional deployment; our work emphasises SLM efficiency, privacy, and edge/on-campus hosting with mixed structured–unstructured sources.	http://www.dbpia.co.kr/Journal/ArticleDetail/NODE12296068
Retrieval-Augmented Generation-Based Chatbot for Prospective and Current University Students	Luluk Setiawati Hartono et al.	2025	International Journal of Engineering, Science and IT	Combined RAG and fine-tuning using Gemma-2B for student information queries; evaluated using RAGAS metrics.	Relies on model fine-tuning and open-source LLMs; our system avoids fine-tuning, instead using document-grounded SLM responses with institutional precedence rules.	https://ijesty.org/index.php/ijesty/article/view/951
Retrieval-Augmented Generation (RAG) Chatbots for Education: A Survey of Applications	Jakub Swacha, Michał Gracel	2025	Applied Sciences (MDPI)	Surveyed 47 RAG-based educational chatbots, analysing architectures, evaluation methods, and educational use cases.	A survey paper, not a system; does not propose SLM-centric, privacy-preserving, campus-deployed solutions like ours.	https://www.mdpi.com/2076-3417/15/8/4234
Small Language Models (SLMs) Can Still Pack a Punch: A Survey	Shreyas Subramanian	2025	arXiv (cs.CL)	Comprehensive survey showing SLMs (1–8B parameters) can rival LLMs in efficiency and performance across tasks.	General SLM theory and benchmarks; does not apply SLMs to campus information retrieval with RAG and source governance, which is the novelty of our project.	https://arxiv.org/abs/2501.05465

PROPOSED SOLUTION

Develop a Campus Information Retrieval Assistant using:

- Small Language Model (SLM)
- Retrieval-Augmented Generation (RAG)

System Workflow:

- User submits query
- System retrieves relevant data
- Generates accurate, context-based response

Data Priority:

- Institutional PDFs
- FAQs
- External sources (if needed)

CIRA

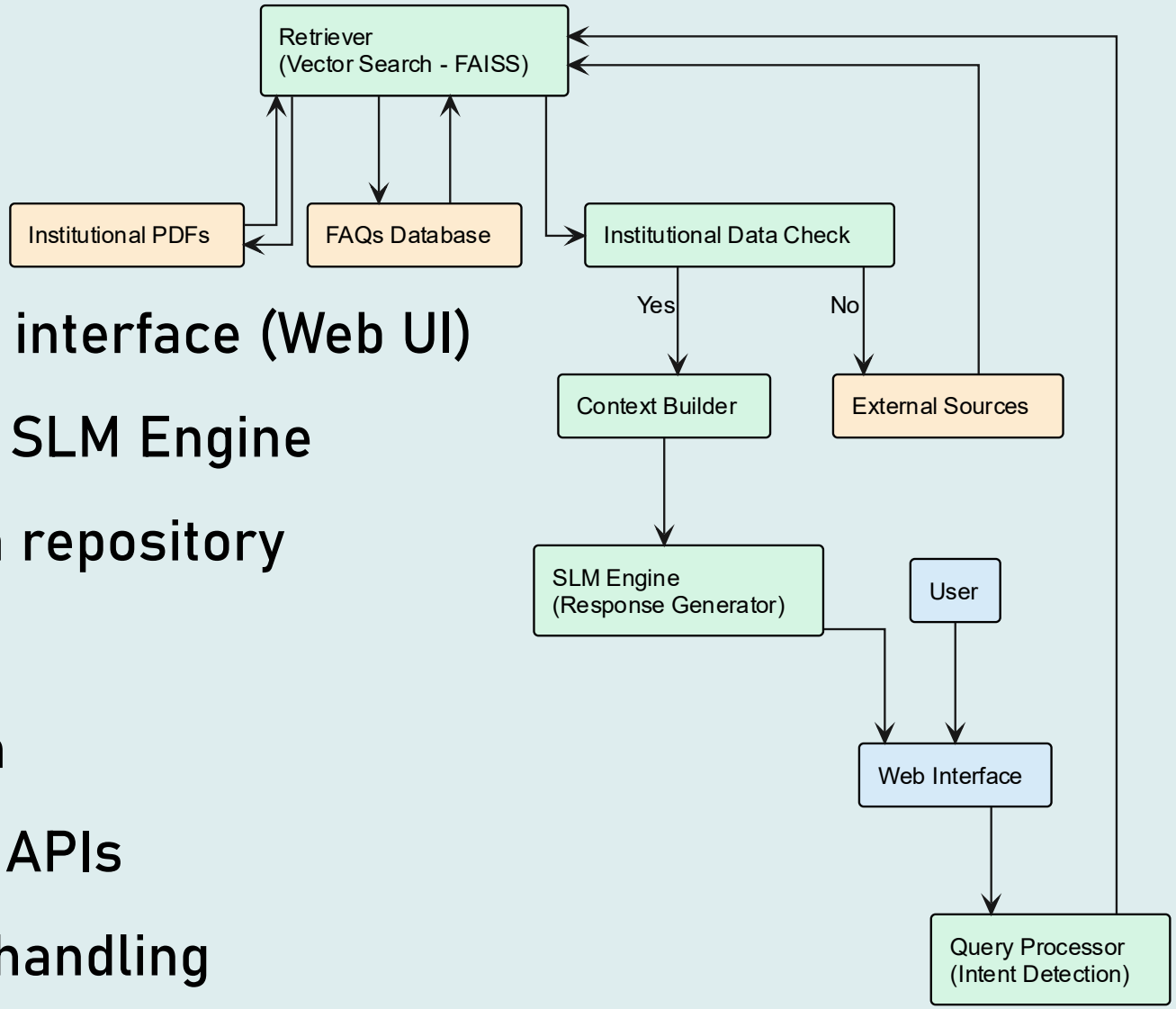
System Architecture

Three-Layer Architecture:

- Presentation Layer: Chat interface (Web UI)
- Processing Layer: RAG + SLM Engine
- Data Layer: Campus data repository

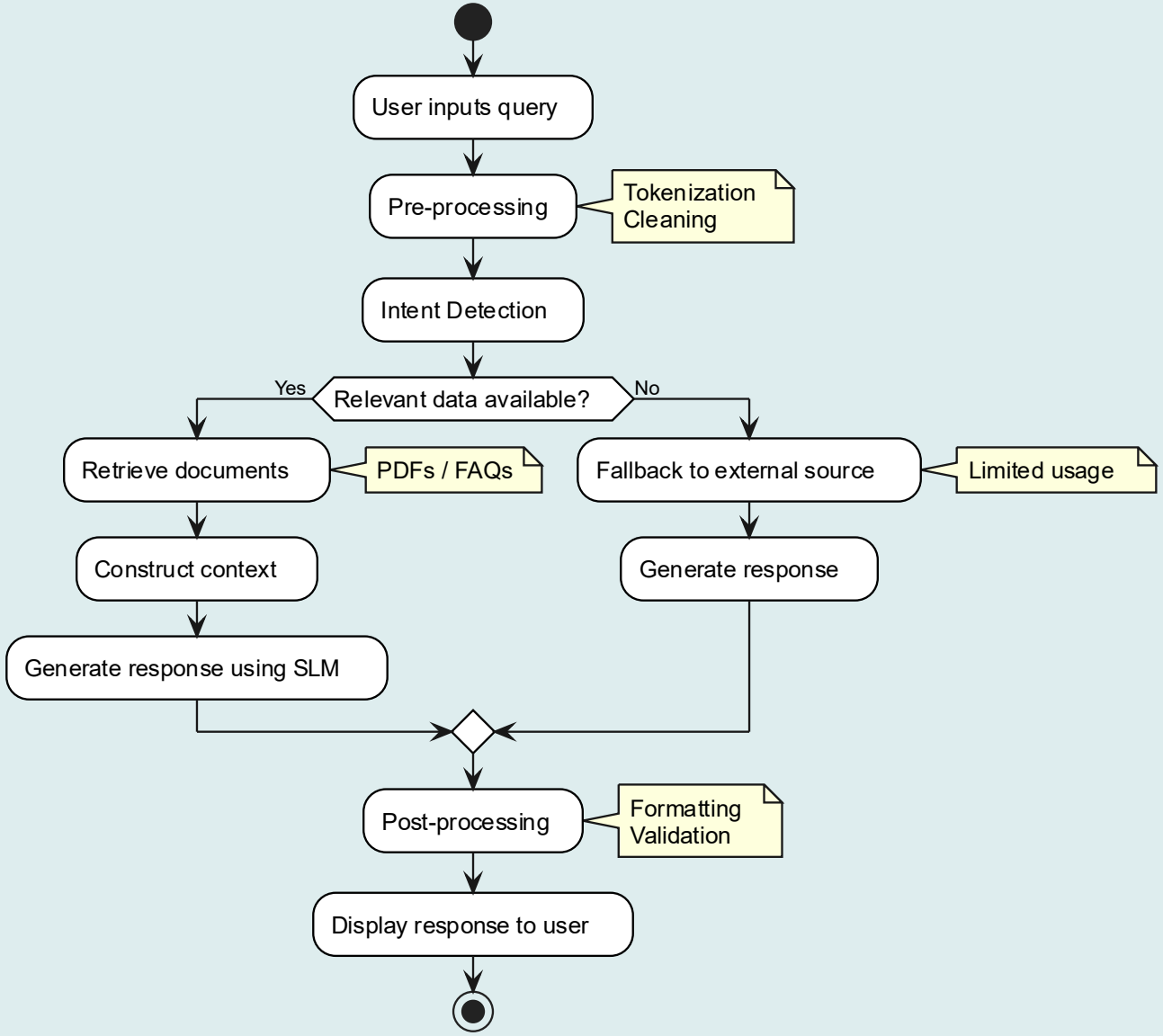
Key Features:

- Fully on-premise system
- No dependency on cloud APIs
- Secure and private data handling



Working Principle

1. User inputs query
2. System understands intent
3. Relevant documents retrieved
4. Context is formed
5. SLM generates final response
6. Response delivered to user



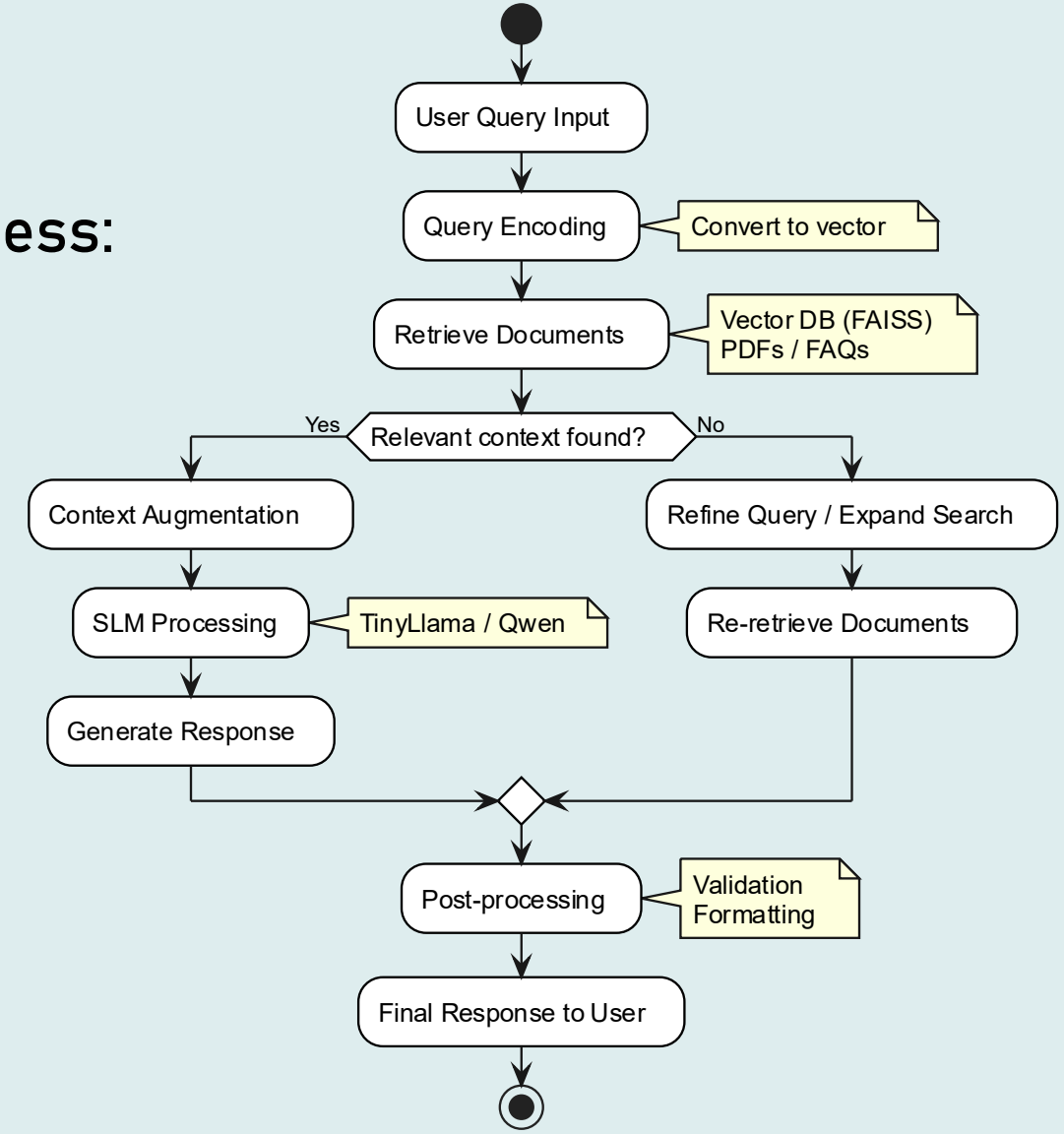
Methodology

Retrieval-Augmented Generation Process:

- Retrieve relevant documents
- Augment query with context
- Generate response using SLM

Advantages:

- Reduces hallucinations
- Ensures factual accuracy
- Uses trusted institutional data



CIRA Inference Pipeline

Retrieval-Augmented Generation with SLM Evaluation

RAG Pipeline

INGESTION
User query

PROCESSING
Text preprocessing
Vector embedding

RETRIEVAL
Hybrid search
RRF fusion

OUTPUT
Final context

embed →

← top-K

context →

FAISS Index

STORAGE
Vector index (IVF)

RETRIEVAL
Similarity search
Top-K results

SLM Engine

CORE
Transformer layers
Token generation

EVALUATION
PPL scoring

Cosine Similarity

$$\mathrm{sim}(A, B) = \frac{A \cdot B}{\|A\| \, \|B\|}$$

A = query vector, B = document chunk

Range [−1, 1]; 1 = identical direction

Perplexity

$$\mathrm{PPL}(W) = 2^{-\frac{1}{N} \sum \log_2 P(w_i)}$$

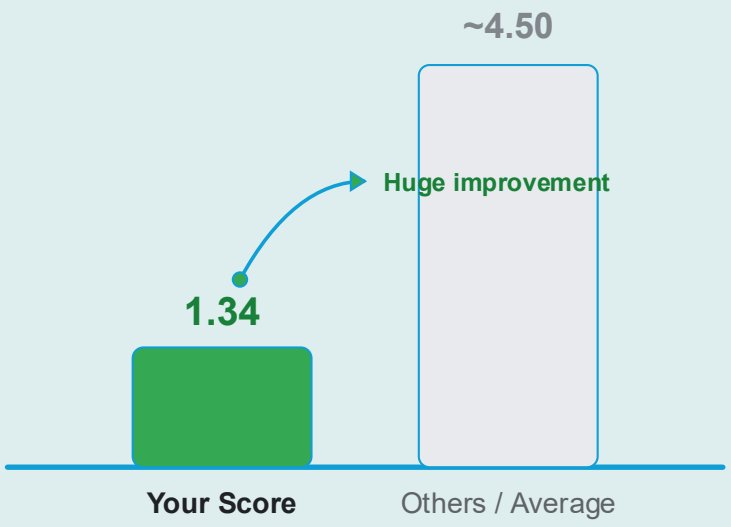
W = generated sequence

N = total token count

$P(w_i)$ = token probability

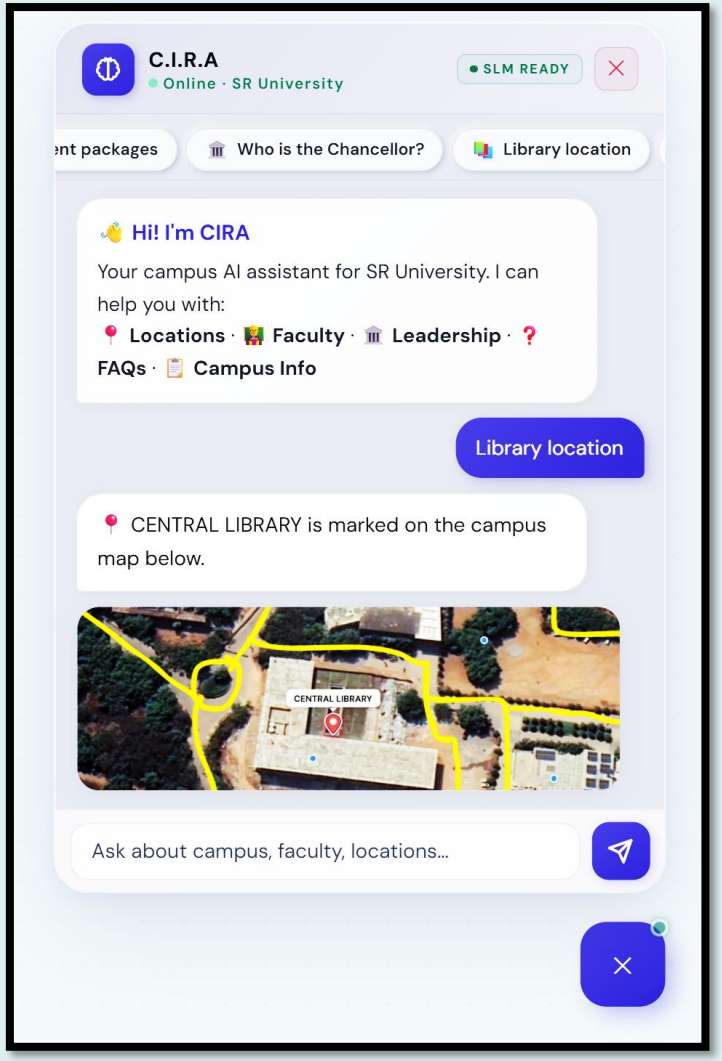
Lower PPL → better fluency

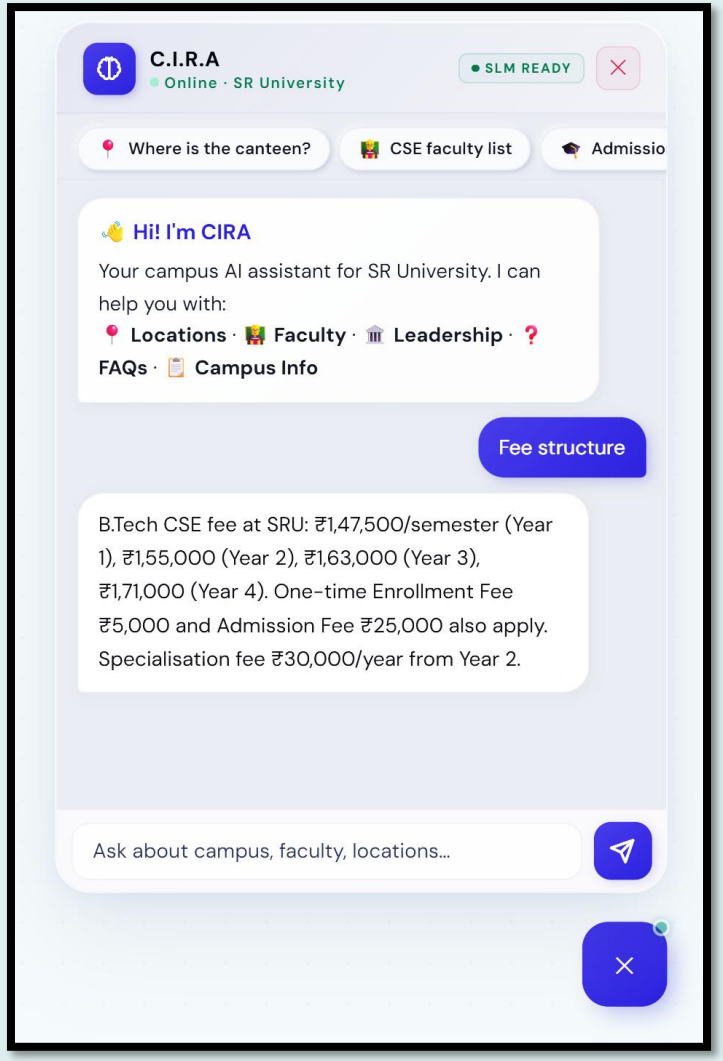
Perplexity Score Comparison

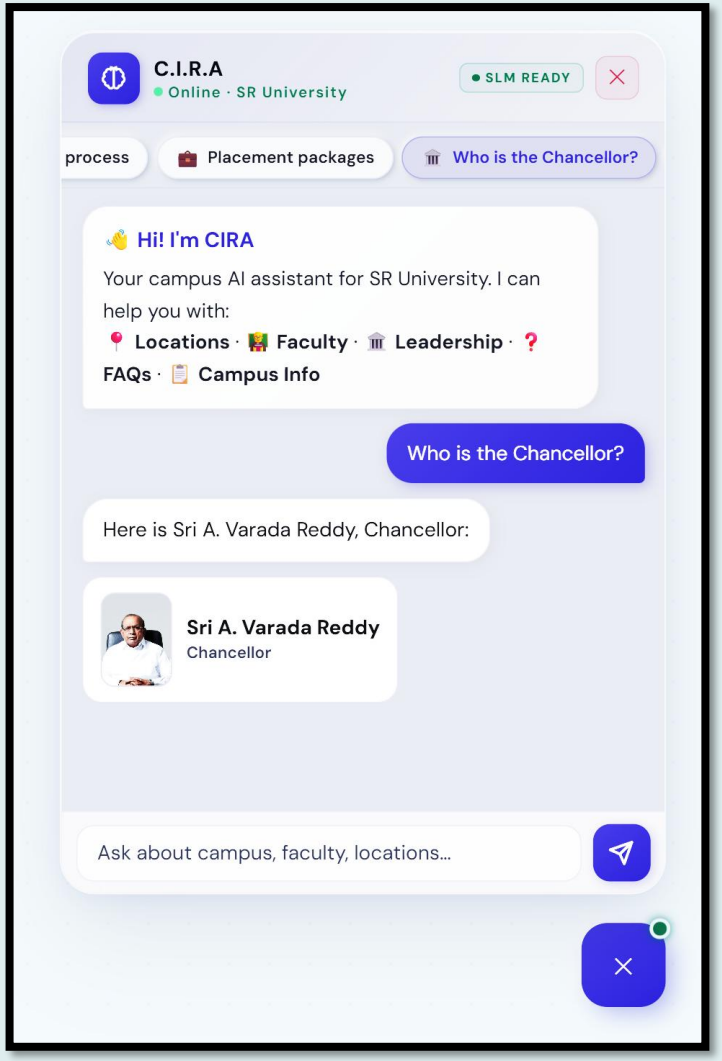


Lower score indicates higher accuracy/confidence

Results


C.I.R.A
Online · SR University
SLM READY
ent packages
Who is the Chancellor?
Library location
Hi! I'm CIRA
Your campus AI assistant for SR University. I can help you with:
Locations · Faculty · Leadership · FAQs · Campus Info
Library location
CENTRAL LIBRARY is marked on the campus map below.
CENTRAL LIBRARY
Ask about campus, faculty, locations...


C.I.R.A
Online · SR University
SLM READY
Where is the canteen?
CSE faculty list
Admissio
Hi! I'm CIRA
Your campus AI assistant for SR University. I can help you with:
Locations · Faculty · Leadership · FAQs · Campus Info
Fee structure
B.Tech CSE fee at SRU: ₹1,47,500/semester (Year 1), ₹1,55,000 (Year 2), ₹1,63,000 (Year 3), ₹1,71,000 (Year 4). One-time Enrollment Fee ₹5,000 and Admission Fee ₹25,000 also apply. Specialisation fee ₹30,000/year from Year 2.
Ask about campus, faculty, locations...


C.I.R.A
Online · SR University
SLM READY
process
Placement packages
Who is the Chancellor?
Hi! I'm CIRA
Your campus AI assistant for SR University. I can help you with:
Locations · Faculty · Leadership · FAQs · Campus Info
Who is the Chancellor?
Here is Sri A. Varada Reddy, Chancellor:
Sri A. Varada Reddy
Chancellor
Ask about campus, faculty, locations...

Results & Advantages

- High accuracy due to grounded responses

- Zero cloud dependency ensures privacy

- Reduced operational cost

- Faster response time

- Scalable for university-wide deployment

Conclusion

- Successfully developed a campus-specific intelligent assistant
- Provides accurate, reliable, and context-aware responses
- Suitable for real-world deployment

Future Scope:

- Voice-based interaction
- Mobile application integration
- Multi-language support
- Integration with university ERP systems

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